

Project Title:

Netflix Content Insights

1. Introduction

This project focuses on analyzing Netflix's content dataset to uncover patterns in movies and TV shows. The goal is to understand how content is distributed across countries, genres, ratings, and time.

2. Objectives

- Analyze the distribution of Movies vs TV Shows
 - Identify top content-producing countries
 - Explore most common genres
 - Study trends in content addition over time
 - Compare growth of Movies vs TV Shows
 - Examine ratings distribution
 - Analyze duration and season patterns
 - Identify frequently appearing directors and actors
 - Understand regional content preferences
-

3. Dataset Source

- Dataset: Netflix Movies and TV Shows Dataset
- Source: Kaggle
- Common dataset name: *Netflix Titles Dataset*
- File format: CSV
- Key columns:

- type (Movie/TV Show)
 - title
 - director
 - cast
 - country
 - date_added
 - release_year
 - rating
 - duration
 - listed_in (genre)
-

4. Tools and Technologies

- Programming Language: Python
 - Libraries:
 - Pandas (data manipulation)
 - NumPy (numerical operations)
 - Matplotlib & Seaborn (data visualization)
 - Environment:
 - Google Colab
-

5. Project Workflow

Step 1: Data Collection

- Import dataset from Kaggle
- Load using Pandas

Step 2: Data Cleaning

- Handle missing values
- Convert date formats
- Split multi-value columns (country, genre, cast)
- Remove duplicates

Step 3: Data Exploration (EDA)

- Univariate Analysis (single variable)
- Bivariate Analysis (comparison between variables)
- Time-series analysis

Step 4: Data Visualization

- Bar charts (countplots)
- Line graphs (trends over years)
- Histograms (duration distribution)
- Pie charts (optional for proportions)

6. Key Analysis Performed

- Movies vs TV Shows distribution
- Top 10 countries producing content
- Genre frequency analysis
- Year-wise content growth
- Movie vs TV show trends over time
- Ratings distribution
- Director and actor frequency
- Duration and season analysis

- Country-wise content type comparison
-

7. Key Insights

- Movies dominate Netflix content overall
 - USA and India are major contributors
 - Dramas and Comedies are most popular genres
 - Content growth surged after 2015 and peaked around 2018–2020
 - TV shows have grown rapidly in recent years
 - TV-MA and TV-14 ratings are most common
 - Most movies fall within 80–120 minutes
 - Majority of TV shows have 1–2 seasons
-

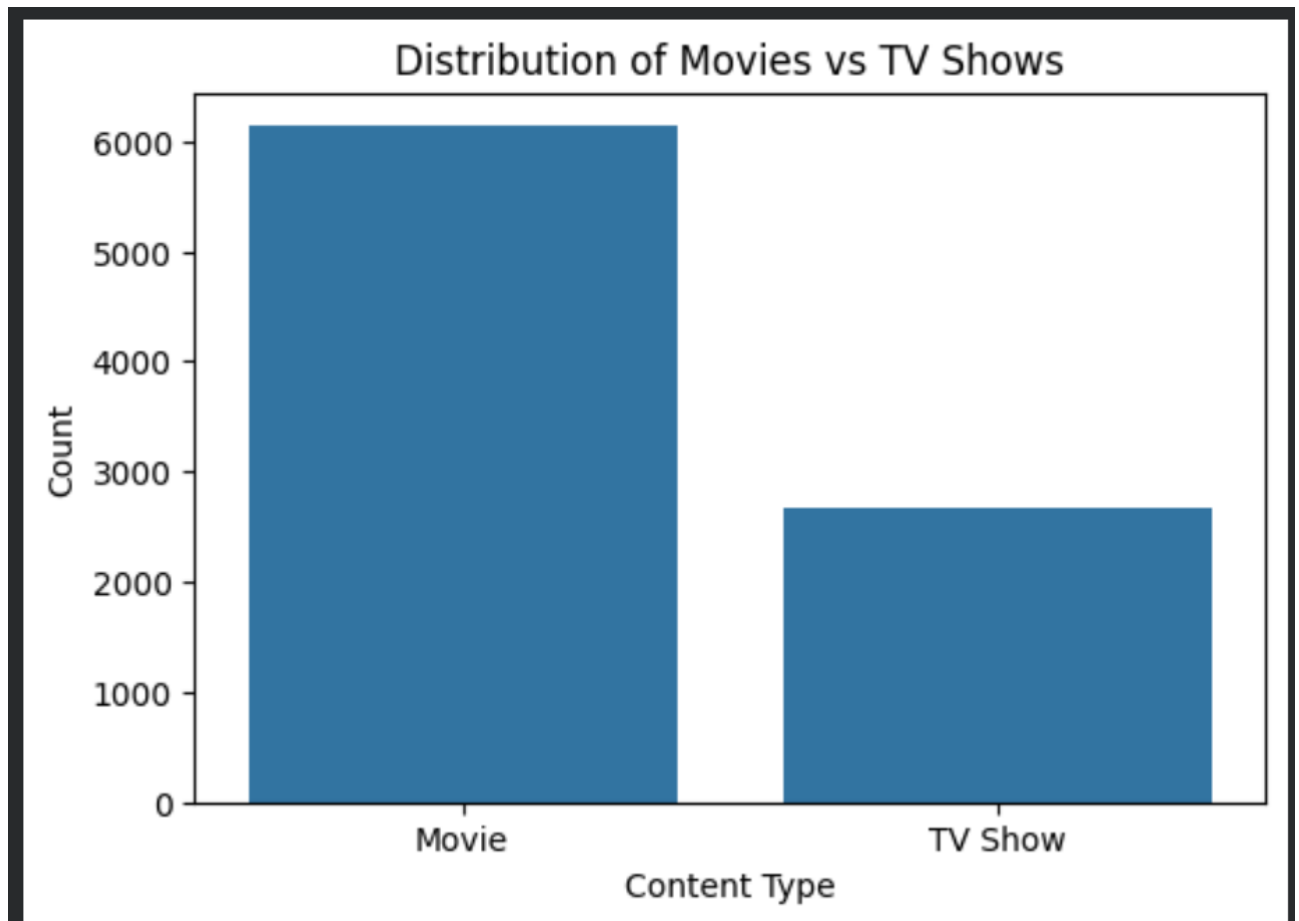
8. Challenges Faced

- Missing data in director, cast, and country fields
 - Multiple values in single columns (comma-separated)
 - Inconsistent formatting in dataset
 - Handling time-based analysis from string dates
-

Exploratory Data Analysis of the Netflix Dataset: Key Questions and Insights

1. What is the distribution of Movies vs TV Shows?

Movies are more in number compared to TV Shows. This shows Netflix has more movie content than series content. It helps understand content type preference on the platform.

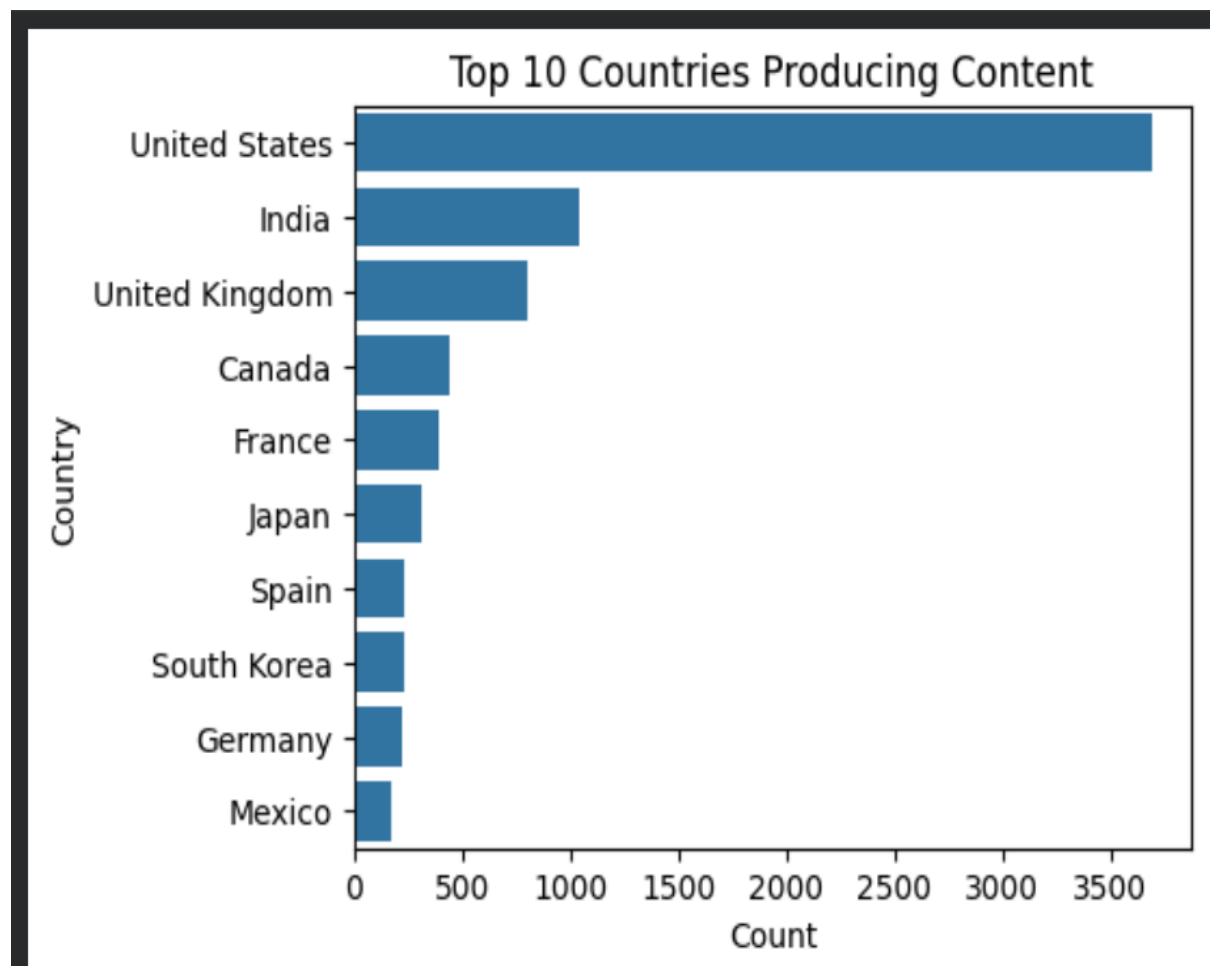


Explanation:

This bar chart compares the number of Movies and TV Shows. Movies appear more frequently, indicating a larger share of film content on Netflix.

2. Which are the top 10 countries producing content?

USA, India, UK, Canada, Japan, and South Korea are among the top contributors. This shows global content distribution on Netflix. It highlights dominance of a few countries.

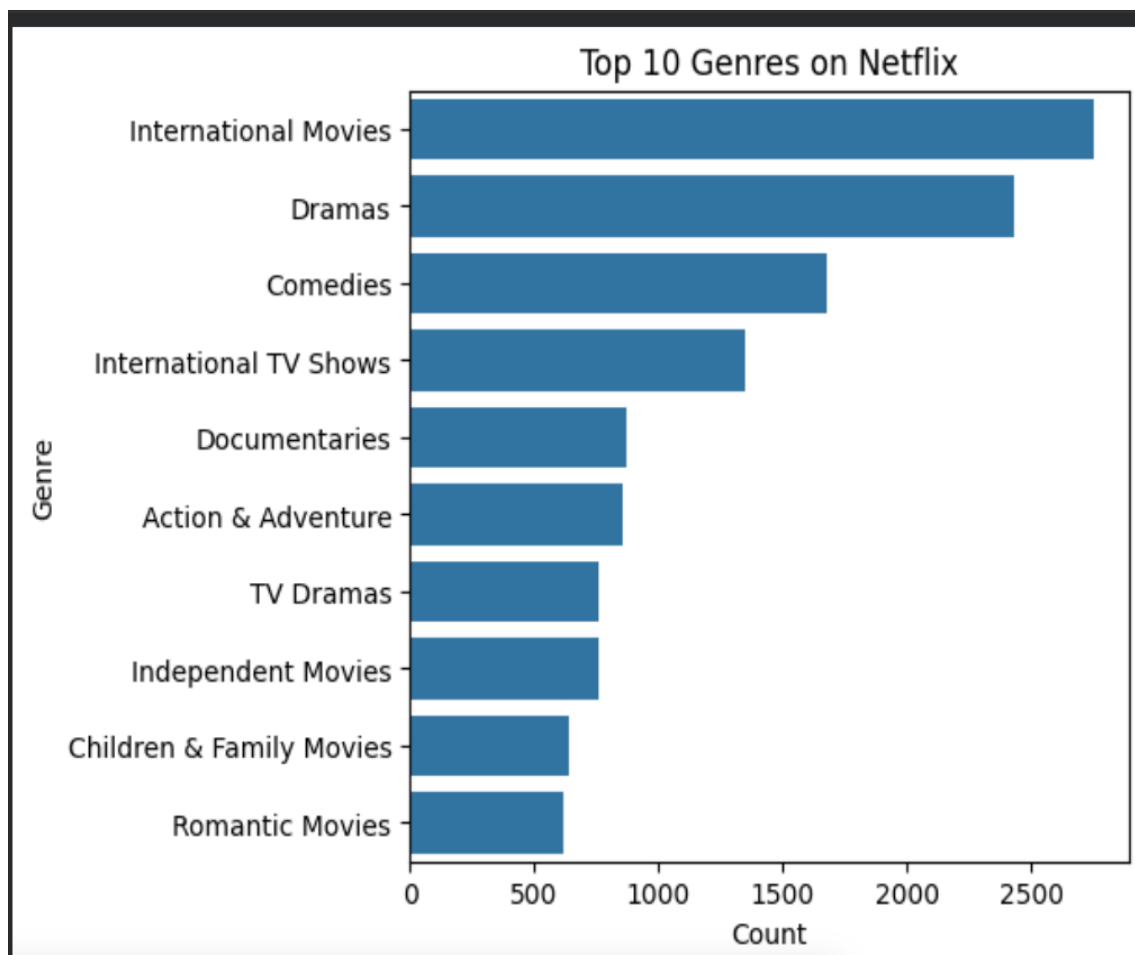


Explanation:

The graph shows the countries with the highest number of titles. Taller bars indicate higher contribution, highlighting dominance of countries like USA and India.

3. What are the most common genres on Netflix?

Dramas, Comedies, and International Movies are most frequent genres. These genres dominate the platform's library. It shows user preference for entertainment types.

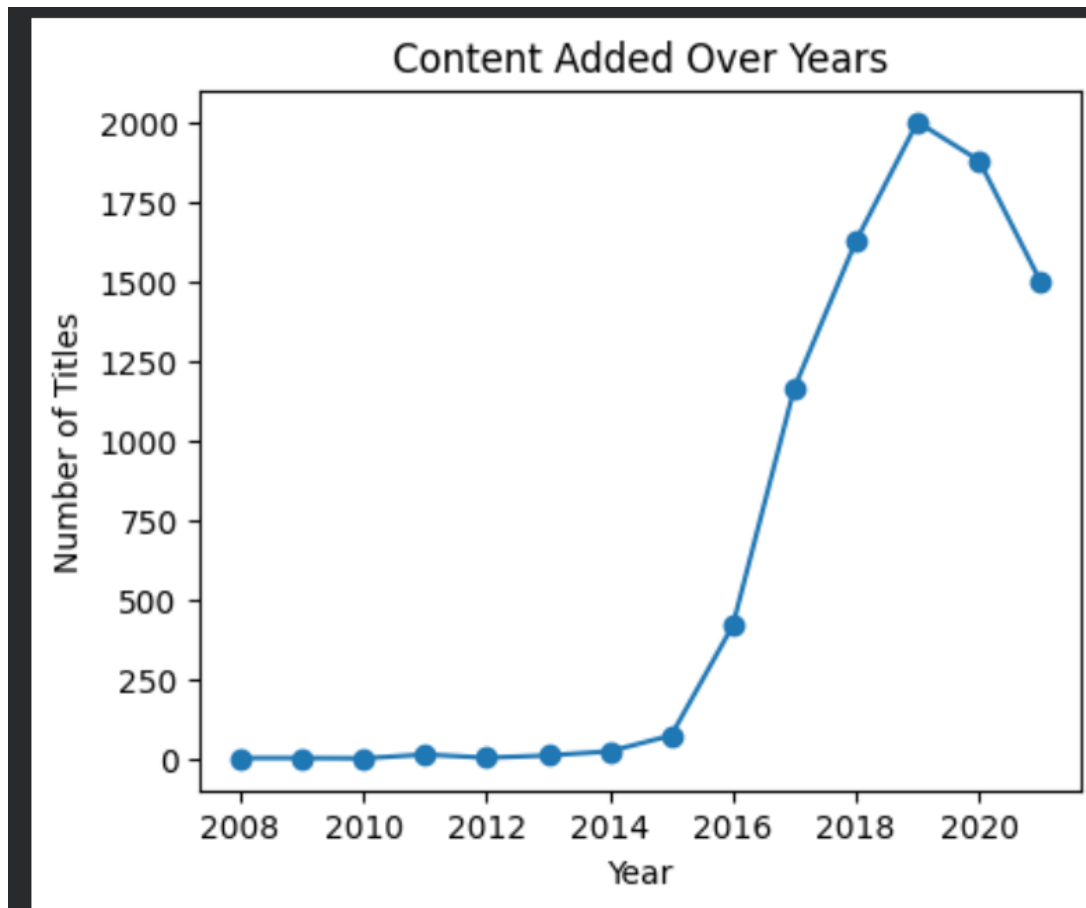


Explanation:

This chart displays the most frequent genres. It shows that Drama and Comedy dominate the platform.

4. How has total content added changed over the years?

Content addition increased significantly after 2015. It peaked around 2018–2020. This reflects Netflix’s rapid expansion phase.

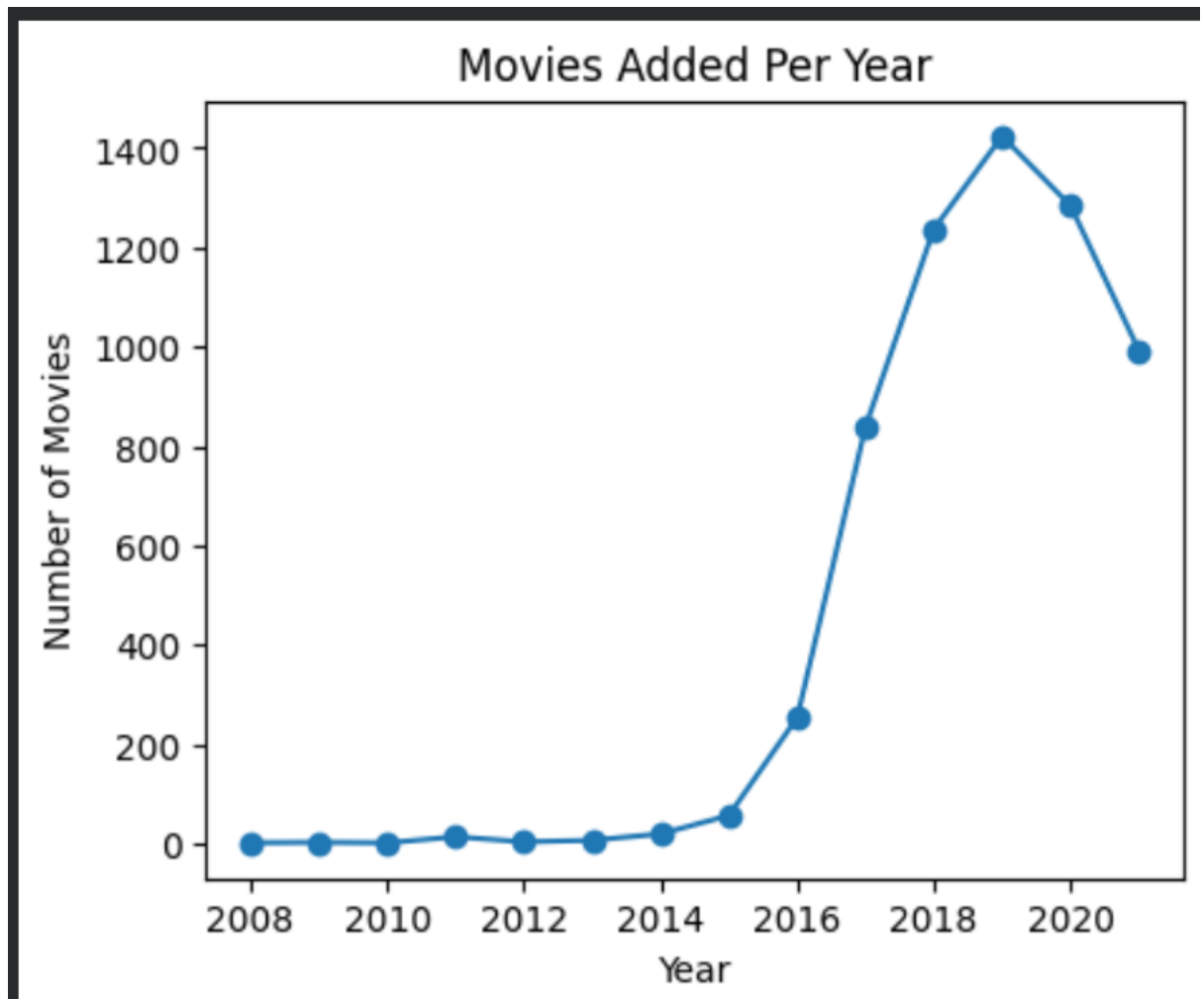


Explanation:

This line graph shows how content additions changed over time. A sharp rise after 2015 indicates rapid platform growth.

5. How many movies were added each year?

Movie additions increased over time with some fluctuations. Growth slowed in recent years. It indicates stabilization of movie content.

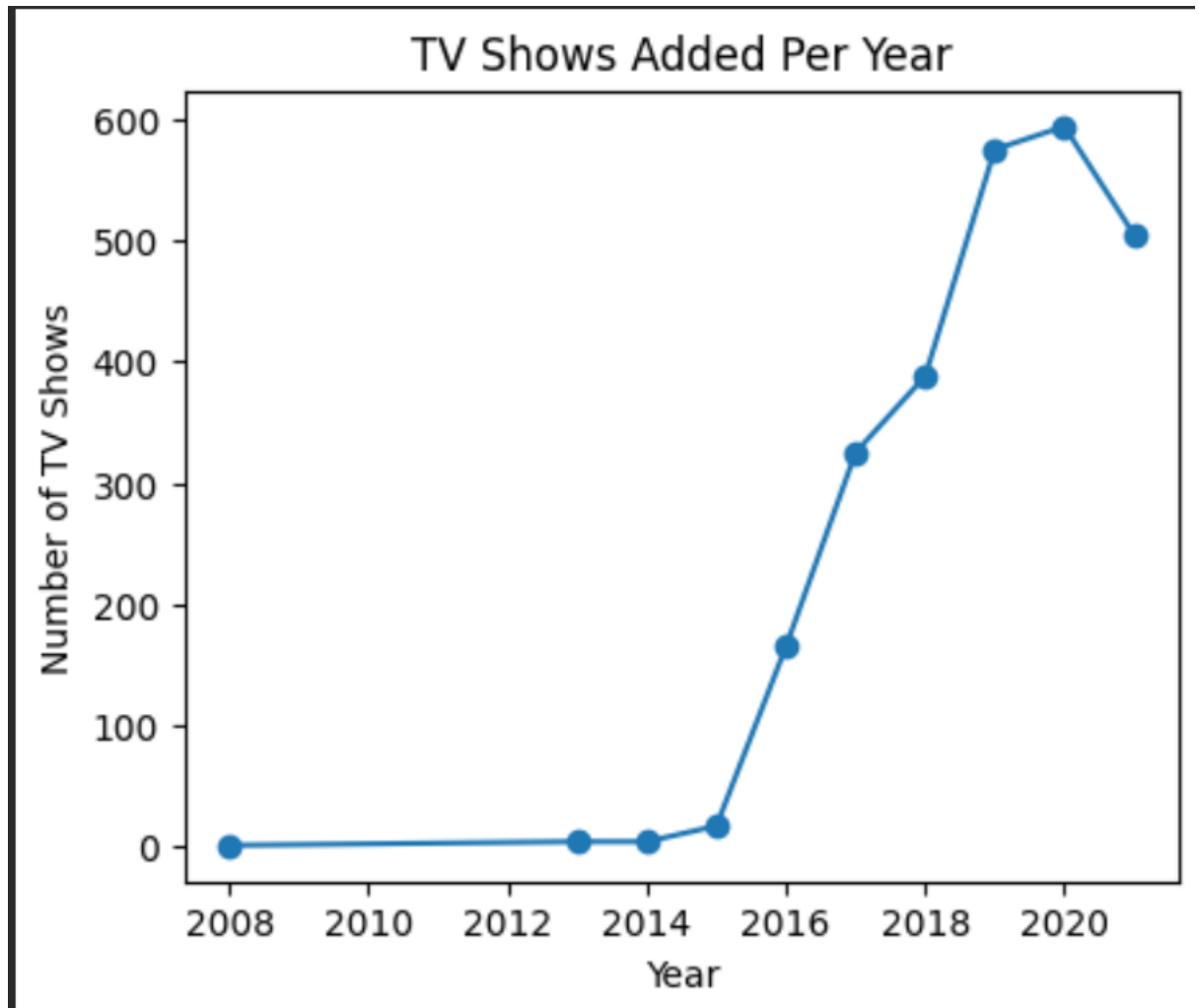


Explanation:

Shows yearly movie additions. Growth increases initially and stabilizes in recent years.

6. How many TV shows were added each year?

TV shows increased steadily, especially after 2017. This shows Netflix's shift toward series content. Growth is more recent compared to movies.

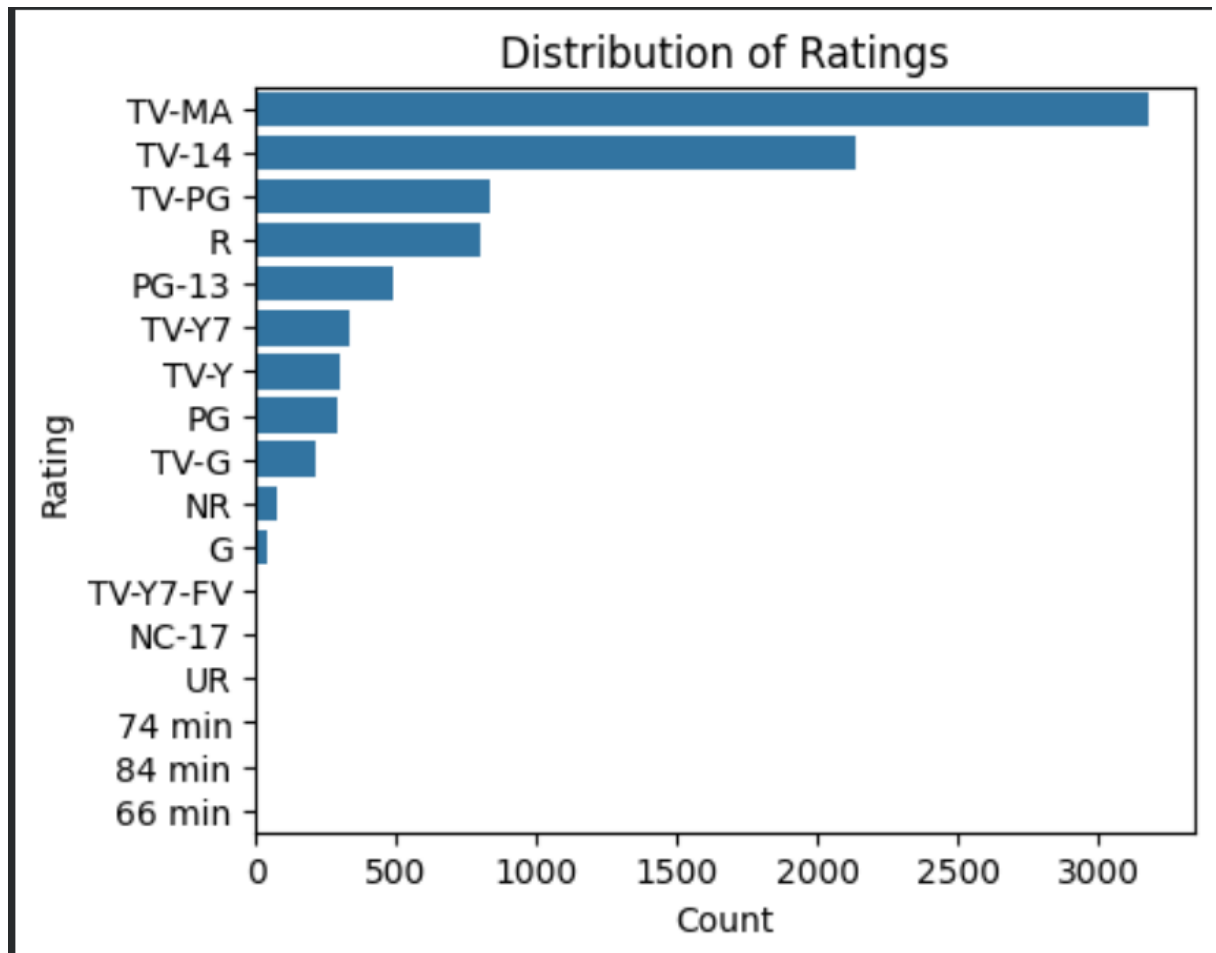


Explanation:

This graph shows a steady increase in TV shows, especially after 2017, indicating a shift toward series content.

7. What is the distribution of content ratings?

TV-MA and TV-14 are the most common ratings. This indicates focus on mature audiences. Family-friendly content is less dominant.

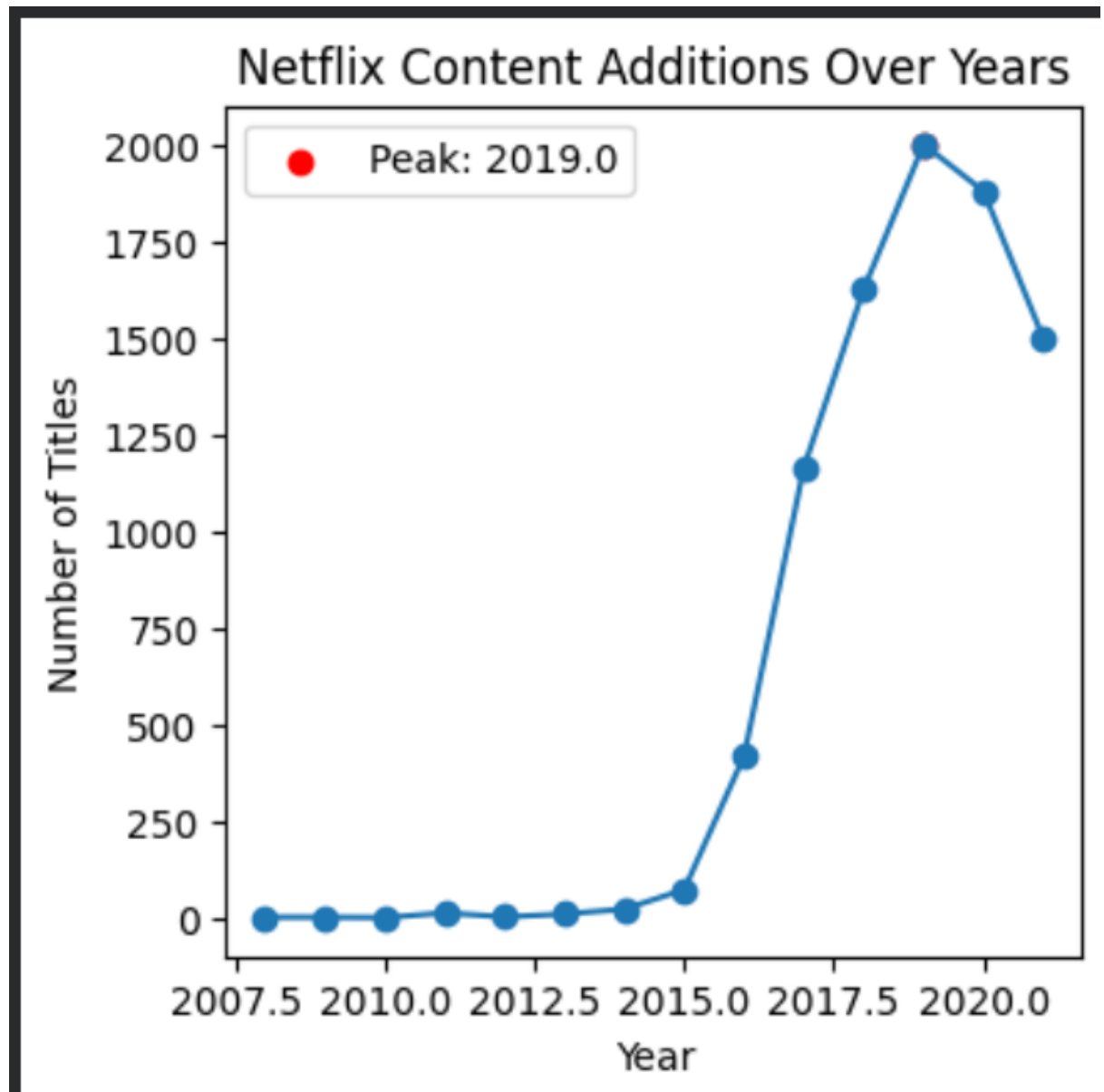


Explanation:

Displays how content is distributed across ratings. TV-MA and TV-14 dominate, showing focus on mature audiences.

8. Which year had the highest content additions?

2018 or 2019 generally shows peak additions. This was Netflix's strongest expansion period. Content growth slows after this peak.

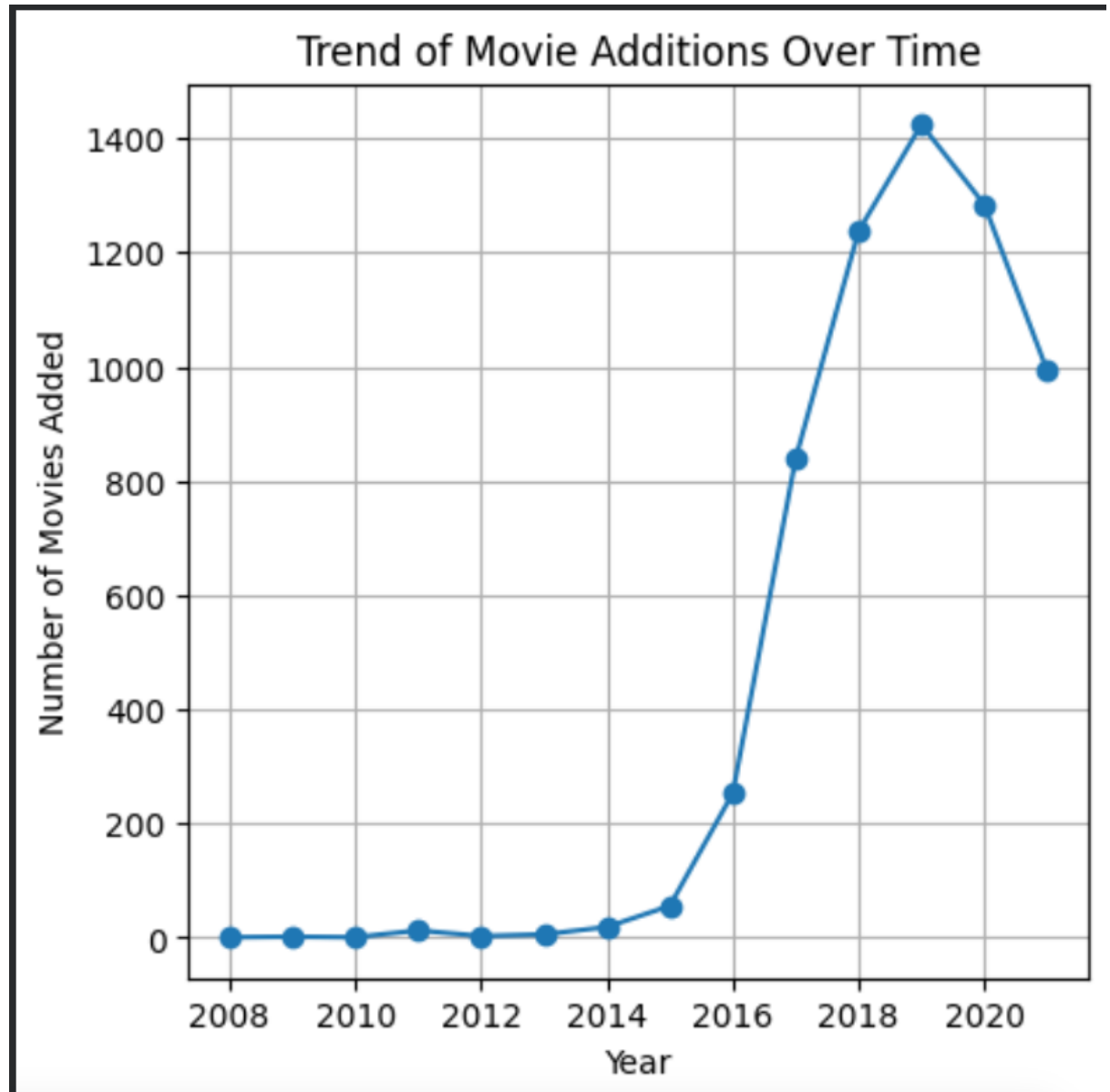


Explanation

This line graph shows the number of Netflix titles added each year. The peak point (highlighted in red) represents the year with the highest content additions, which is typically around **2018 or 2019**, showing Netflix's fastest expansion period.

9. What is the trend of movie additions over time?

Movies increased steadily over the years. Peak occurred before recent stabilization. It shows early focus on film content.

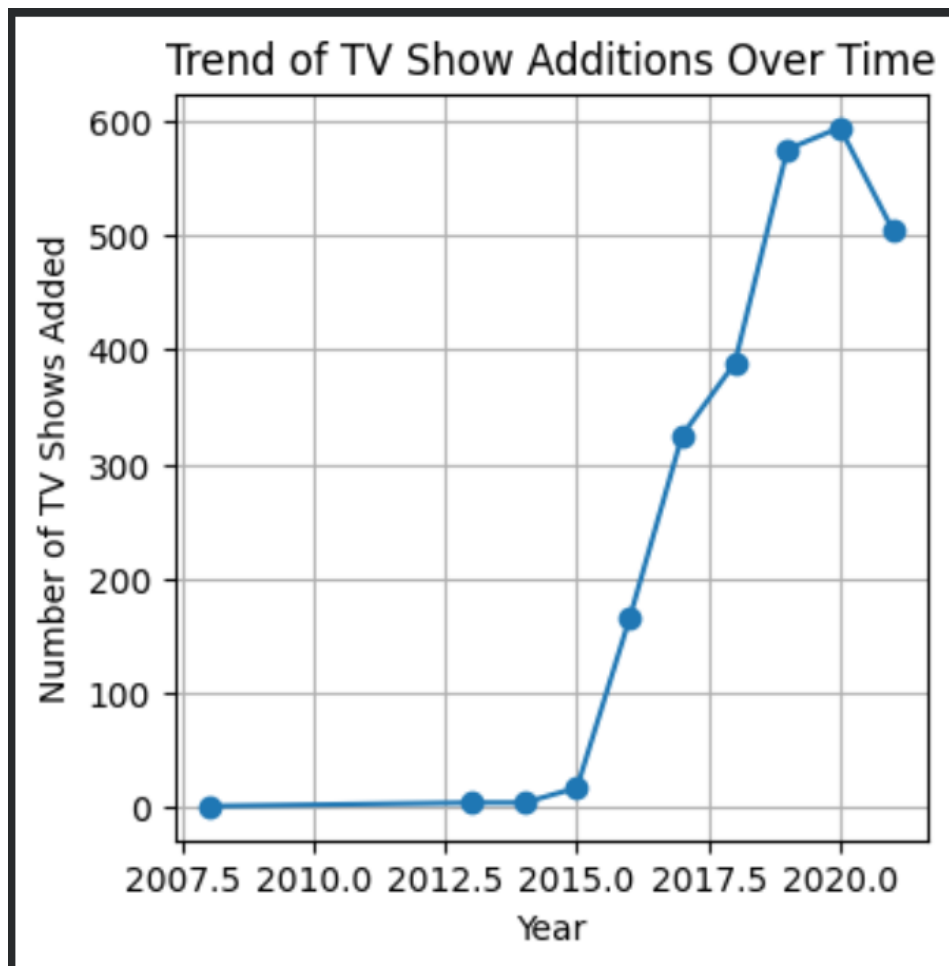


Explanation

This line graph shows how movie additions changed over time on Netflix. The trend increases steadily in earlier years, reaches a peak period, and then stabilizes in recent years. This indicates that Netflix initially focused more on expanding its movie library before balancing it with other content types like TV show.

10.What is the trend of TV show additions over time?

TV shows show continuous growth in recent years. This reflects increased investment in original series. It is becoming a key content strategy.

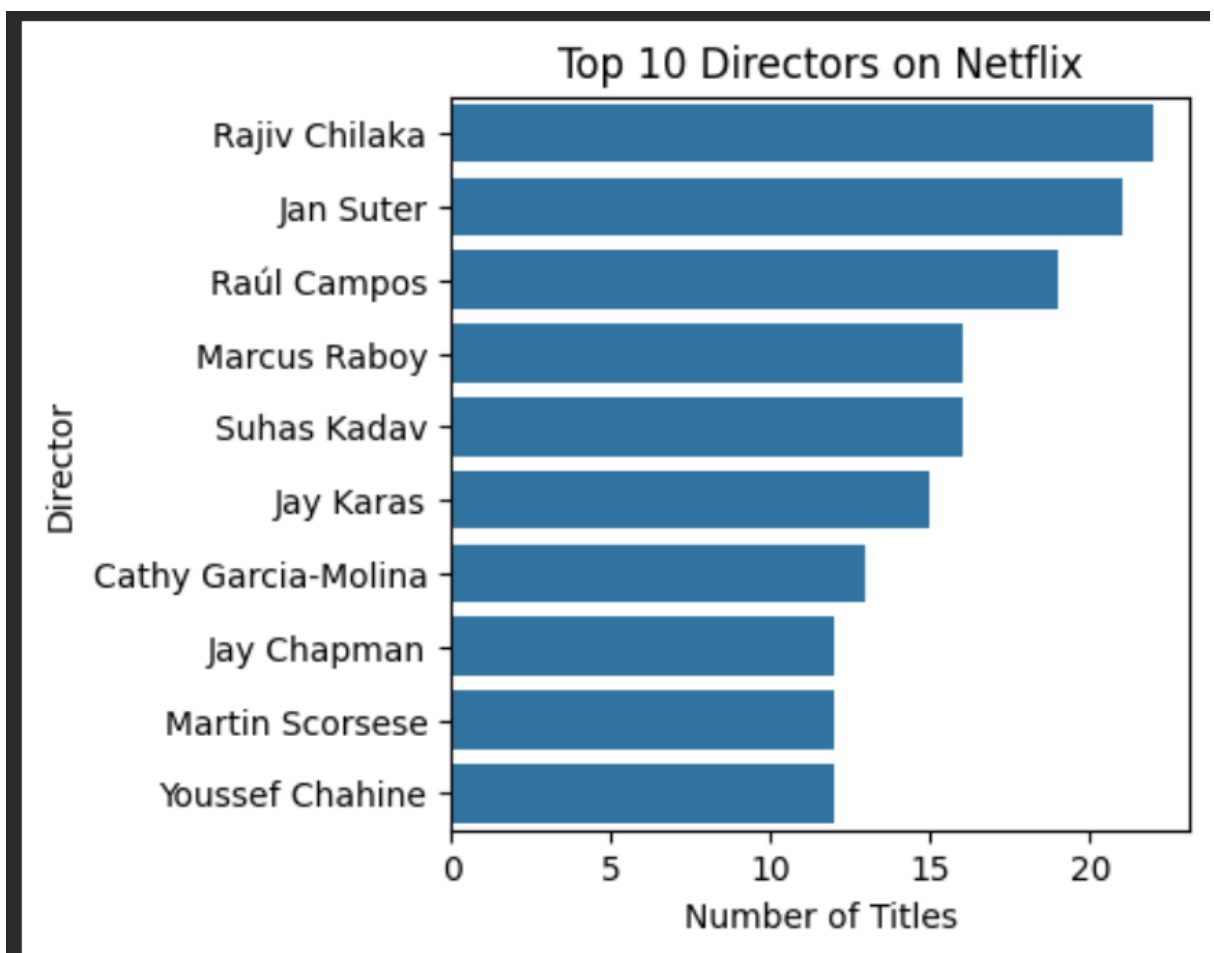


Explanation

This line graph shows the trend of TV show additions over the years on Netflix. The pattern shows slow growth in early years followed by a sharp increase after 2017. This indicates Netflix's strategic shift toward producing and investing more in original series content in recent years.

11. Which directors have the most content?

A few directors contribute multiple titles. Most directors appear only once. This shows high content diversity.

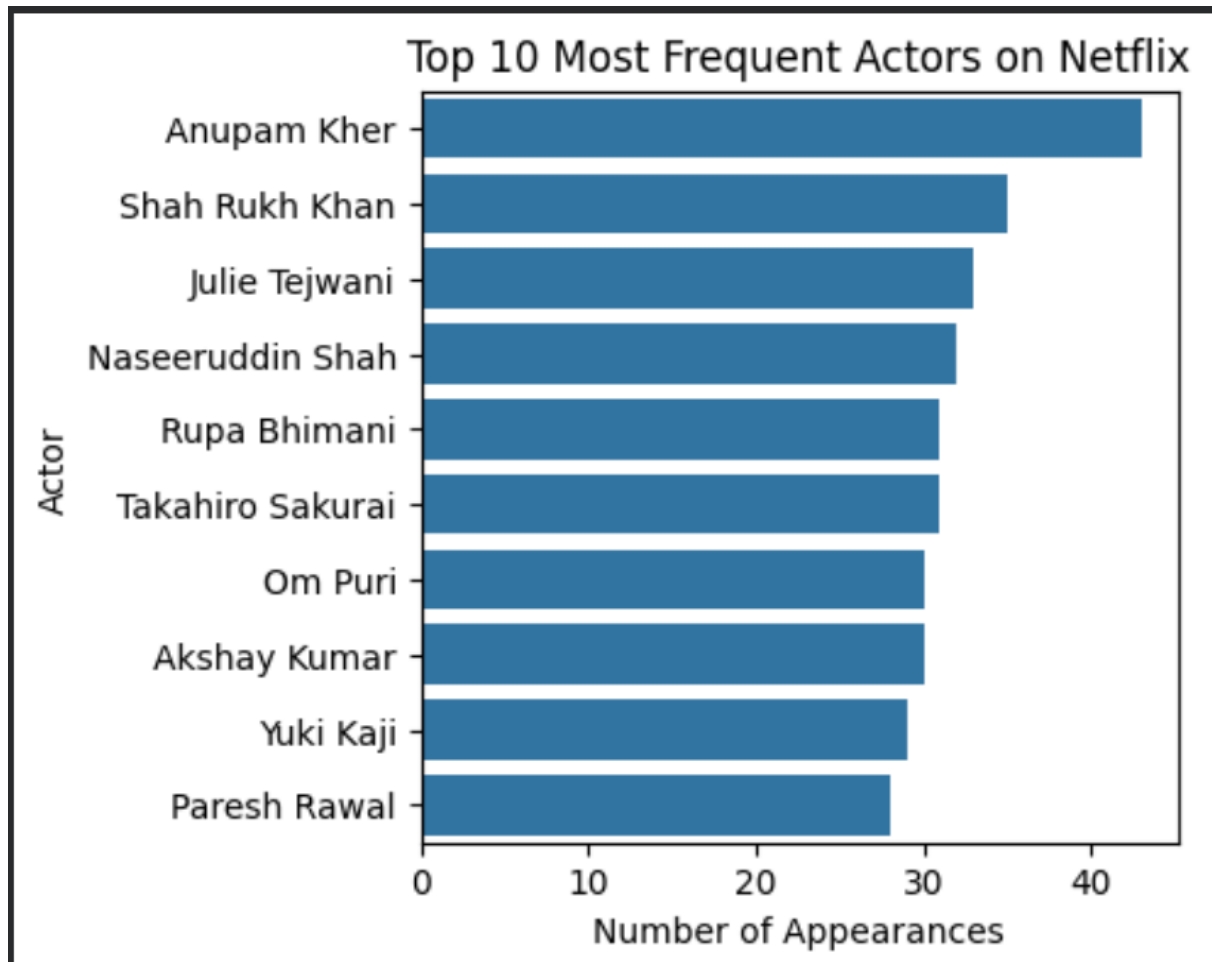


Explanation

This bar chart shows the directors with the highest number of titles on Netflix. A small number of directors contribute multiple shows or movies, while most directors appear only once. This indicates that Netflix has a highly diverse set of creators with limited repetition among directors.

12. Which actors appear most frequently?

Some actors appear repeatedly across titles. USA and India-based actors dominate. It shows repeated collaborations in content.

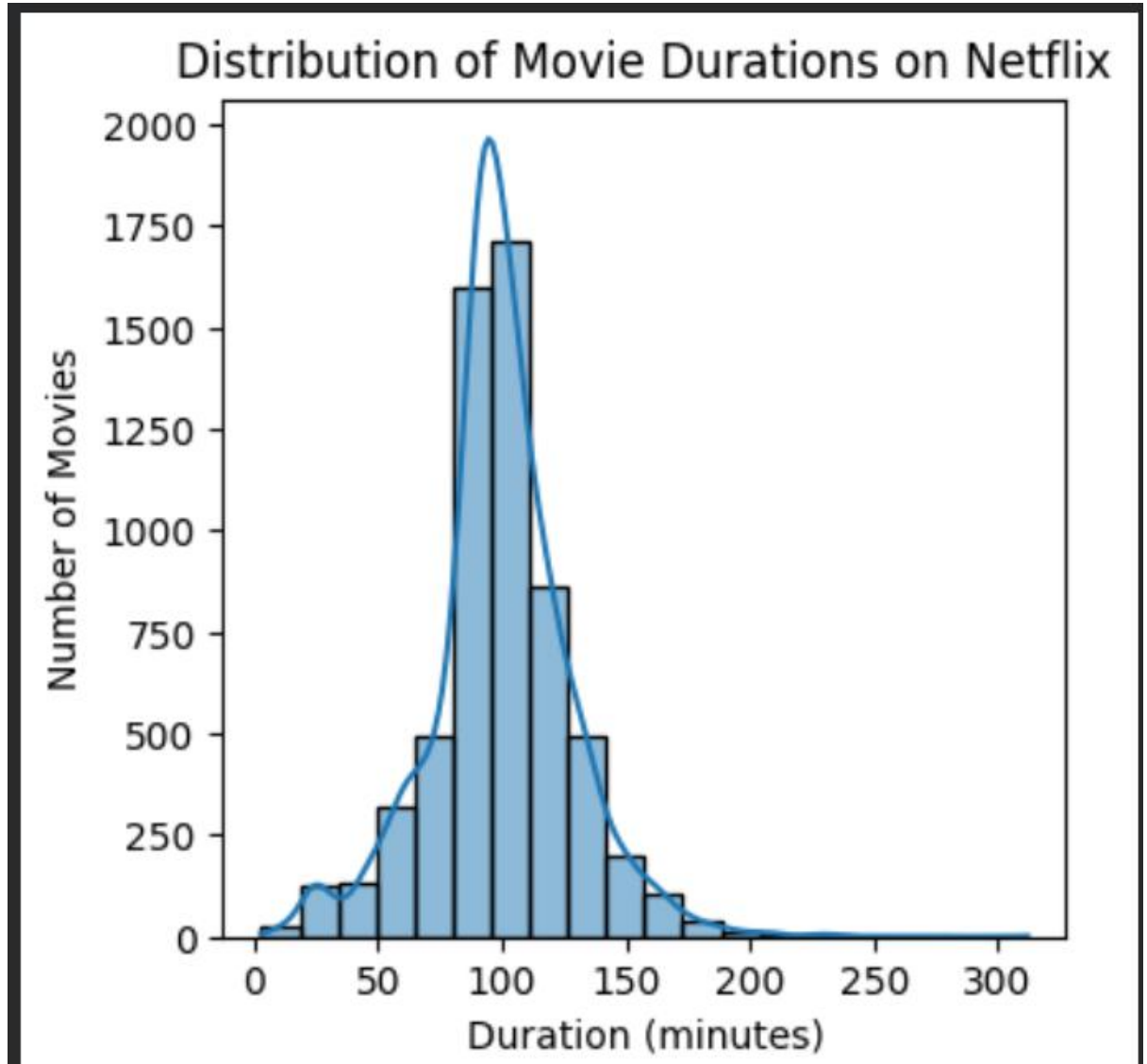


Explanation

This bar chart shows actors who appear most frequently across Netflix titles. A small group of actors appears repeatedly due to multiple collaborations. This indicates that while Netflix has a large cast pool, certain actors are used more often in different productions.

13. What is the distribution of movie durations?

Most movies are between 80–120 minutes. Very short or very long movies are rare. This reflects standard movie length.

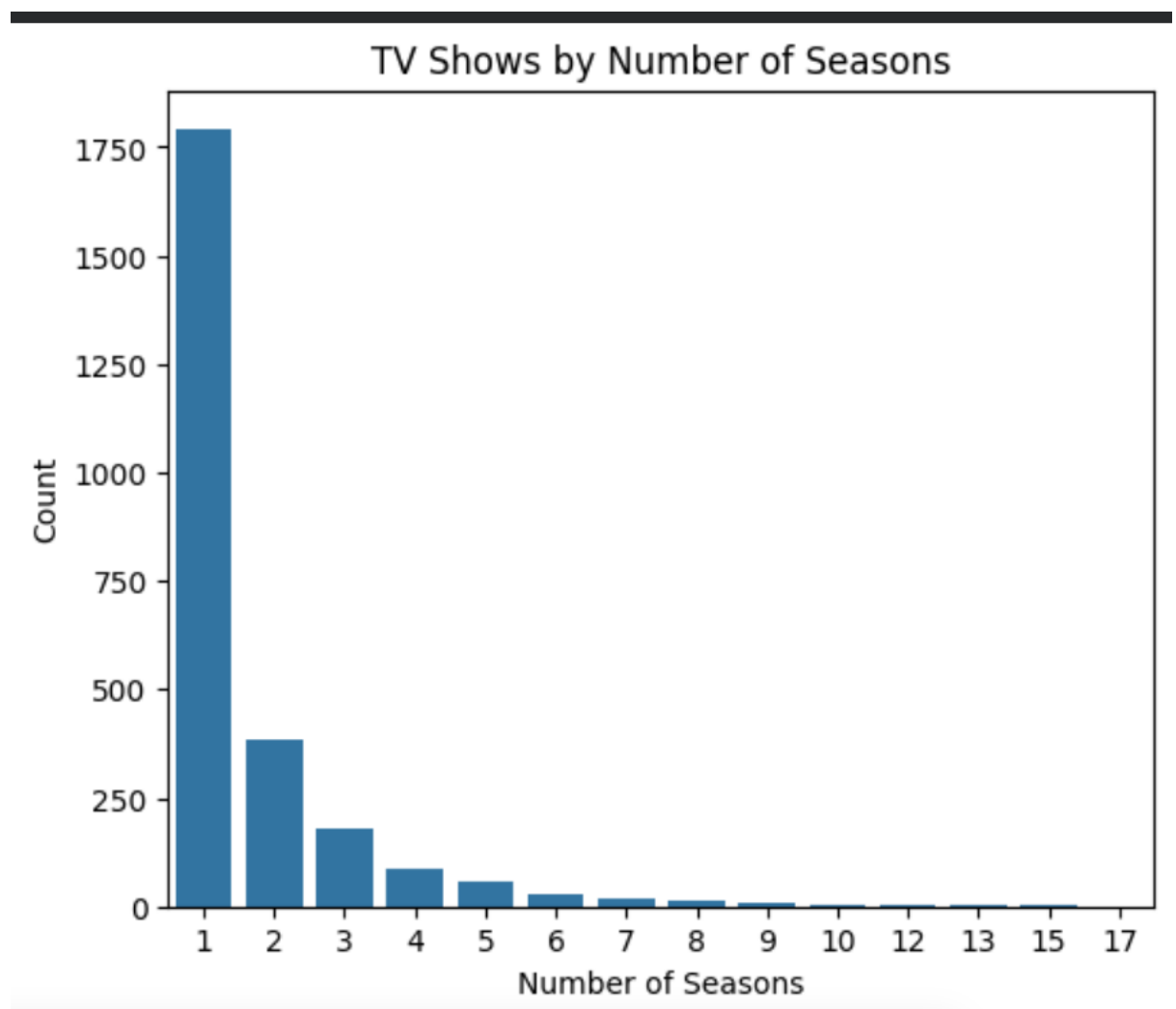


Explanation

This histogram shows how movie durations are distributed on Netflix. Most movies fall between 80–120 minutes, forming a strong central peak. Very short or very long movies are rare, indicating that Netflix content mostly follows standard feature-length formats.

14. How many TV shows have multiple seasons?

Most TV shows have 1–2 seasons. Few shows extend beyond that. This indicates short-series trend on Netflix.

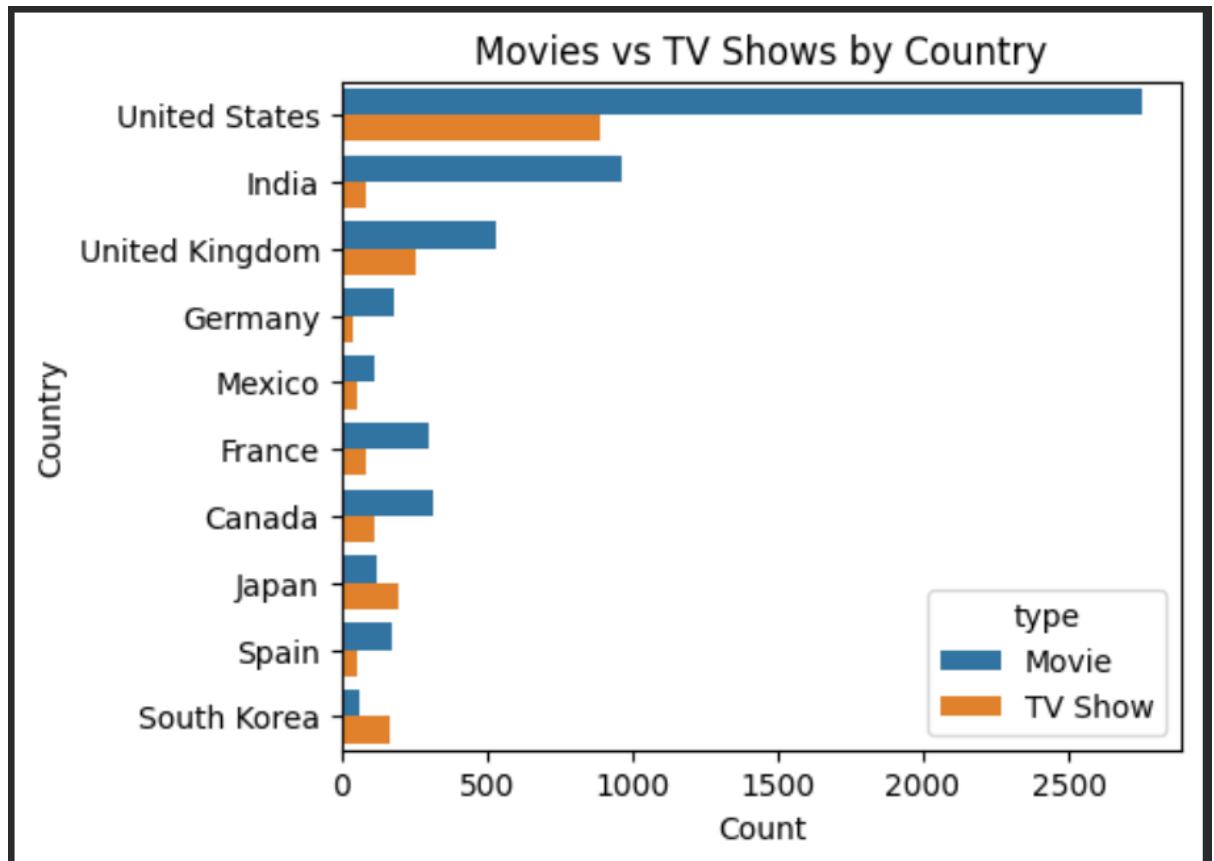


Explanation

This bar chart shows how TV shows are distributed based on the number of seasons. Most shows have 1–2 seasons, while only a few extend beyond that. This reflects Netflix’s preference for short-series formats and limited-season storytelling.¹

15.Which countries produce mostly Movies vs TV Shows?

USA leads in both categories. India produces more movies than TV shows. This shows regional content focus differences.

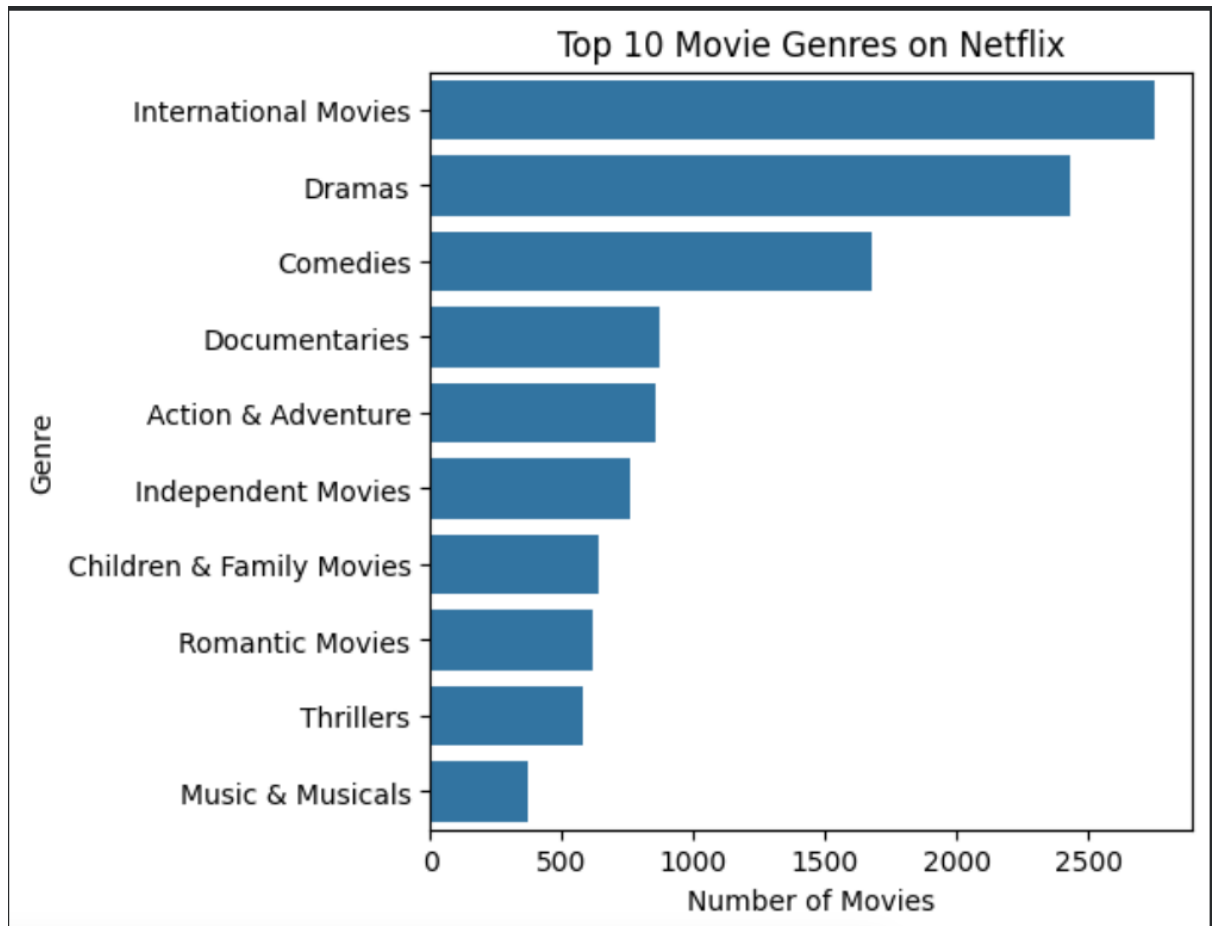


Explanation

This graph compares the number of Movies and TV Shows produced by top countries. The USA leads in both categories, showing strong dominance in content production. India produces significantly more movies than TV shows, indicating a stronger focus on films compared to series in its content contribution.

16. Top 10 genres for movies?

Dramas, Action, and Comedies dominate movie genres. These are widely preferred globally. They form the core movie library.

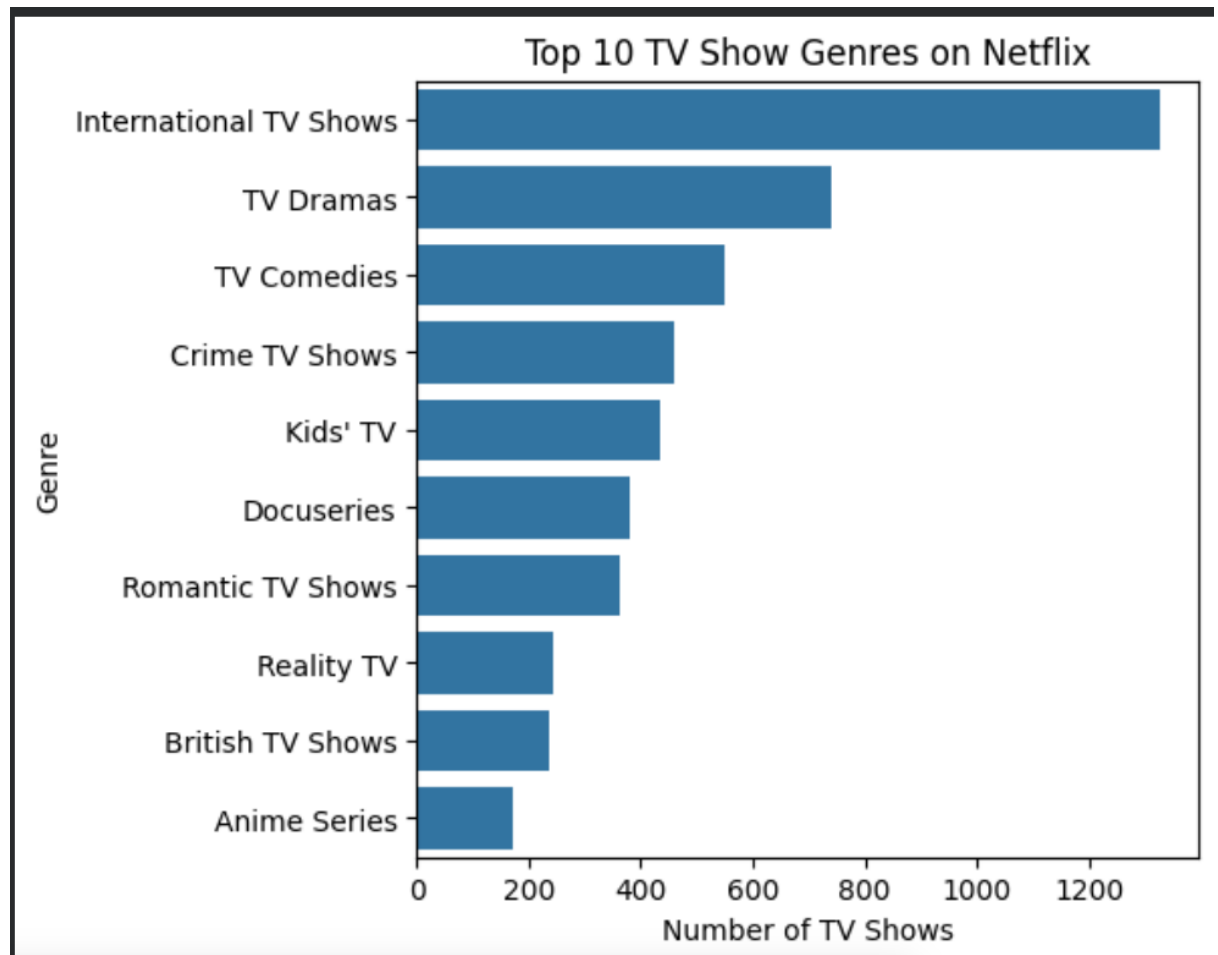


Explanation

This bar chart shows the most common genres in Netflix movies. Dramas, Action, and Comedies dominate the list, indicating strong global audience preference for these categories. These genres form the core of Netflix's movie content library.

17. Top 10 genres for TV shows?

Dramas, Crime, and International TV Shows are most common. These genres attract long-term viewers. They dominate series content.

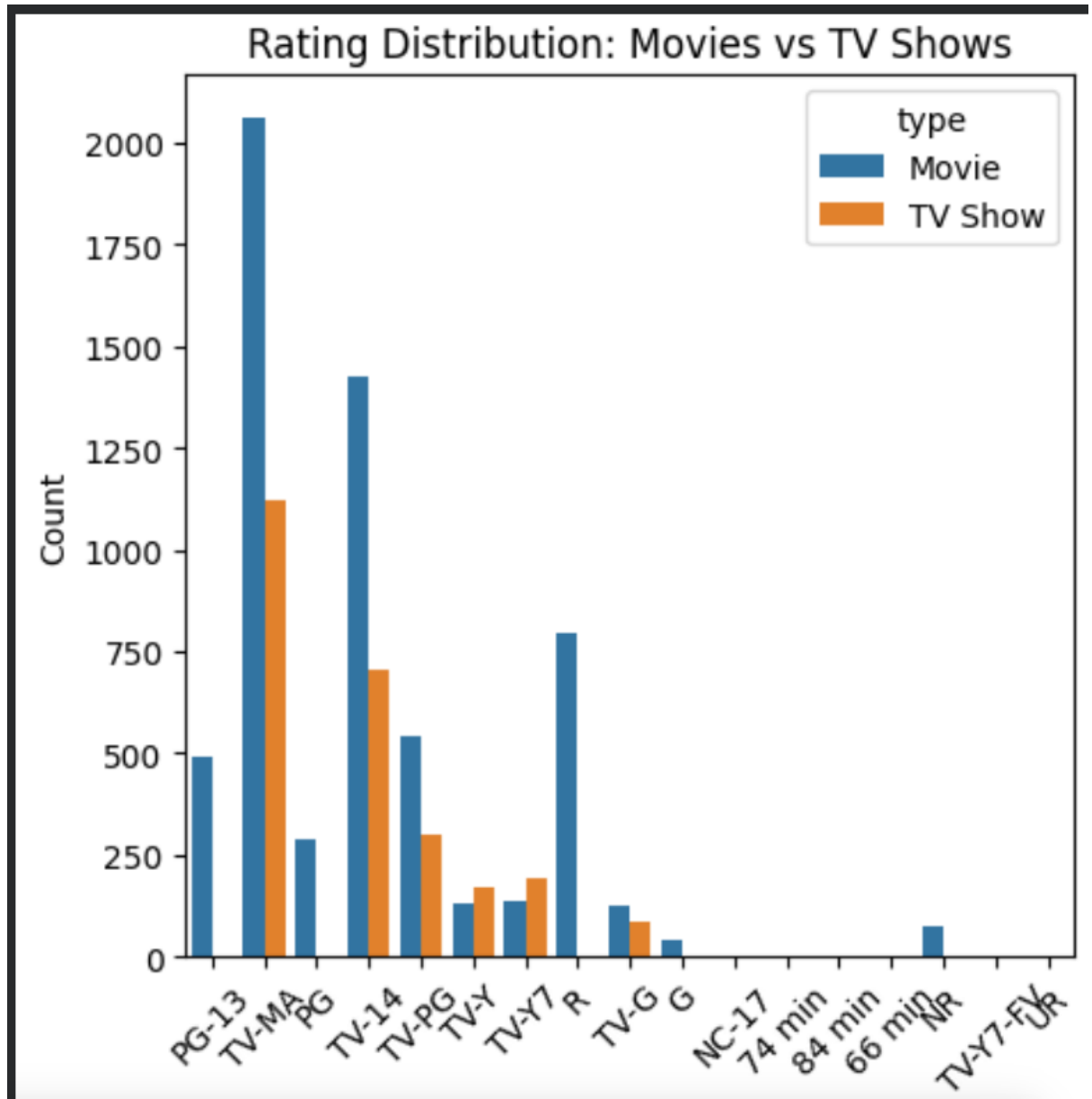


Explanation

This bar chart shows the most frequent genres in Netflix TV shows. Dramas, Crime, and International TV Shows are the most common categories. These genres are preferred for long-form storytelling and help attract sustained viewer engagement over multiple episodes.

18. How does rating differ between Movies and TV Shows?

Movies mostly have TV-MA ratings. TV Shows include more TV-14 and TV-PG. This shows difference in audience targeting.

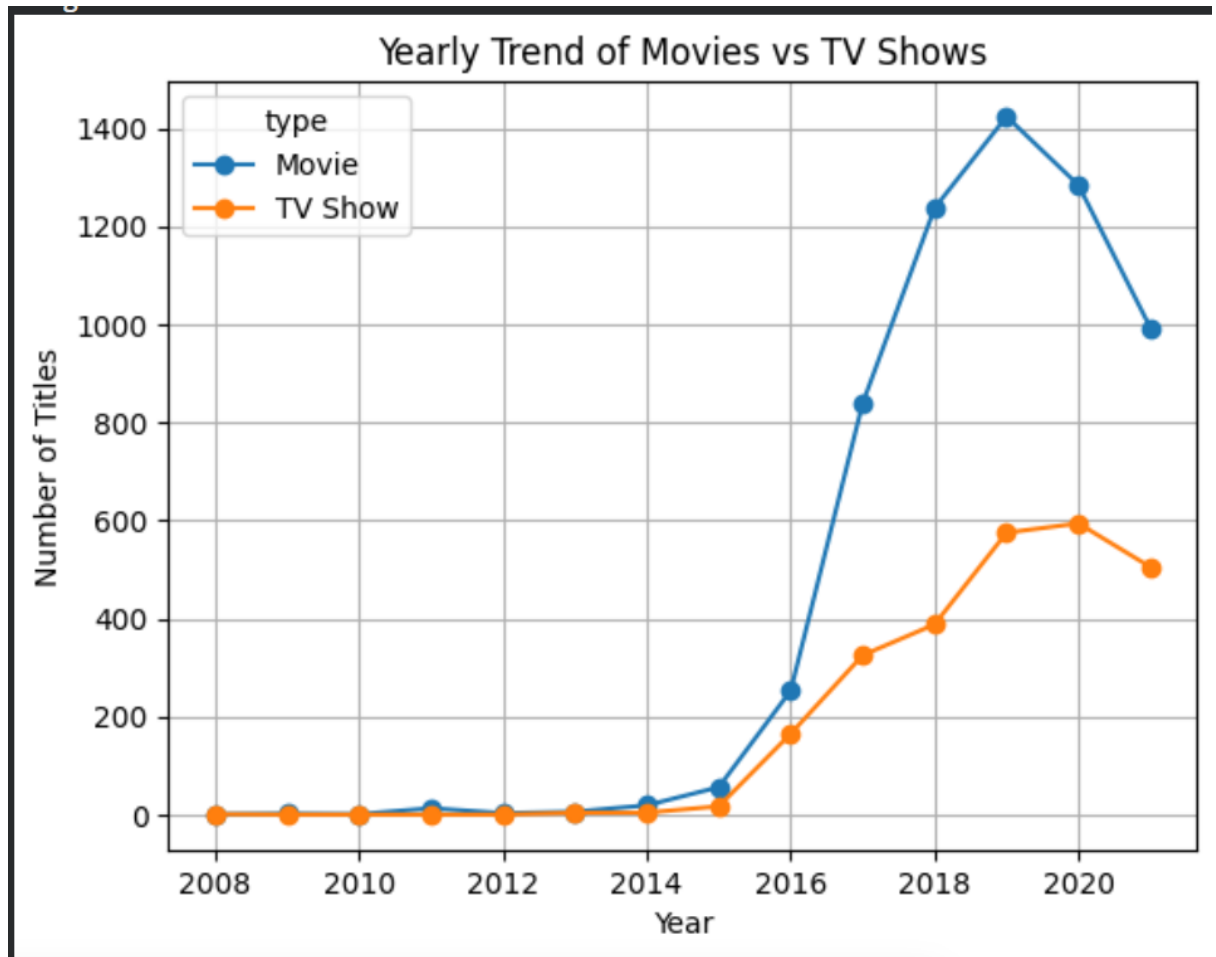


Explanation

This grouped bar chart compares content ratings between Movies and TV Shows. Movies are mostly concentrated in TV-MA ratings, indicating mature audience targeting. TV Shows are more spread across TV-14 and TV-PG, showing a relatively broader audience range compared to movies.

19. Yearly trend of content types?

Movies dominate earlier years. TV shows increase significantly after 2017. This shows shift in content strategy.

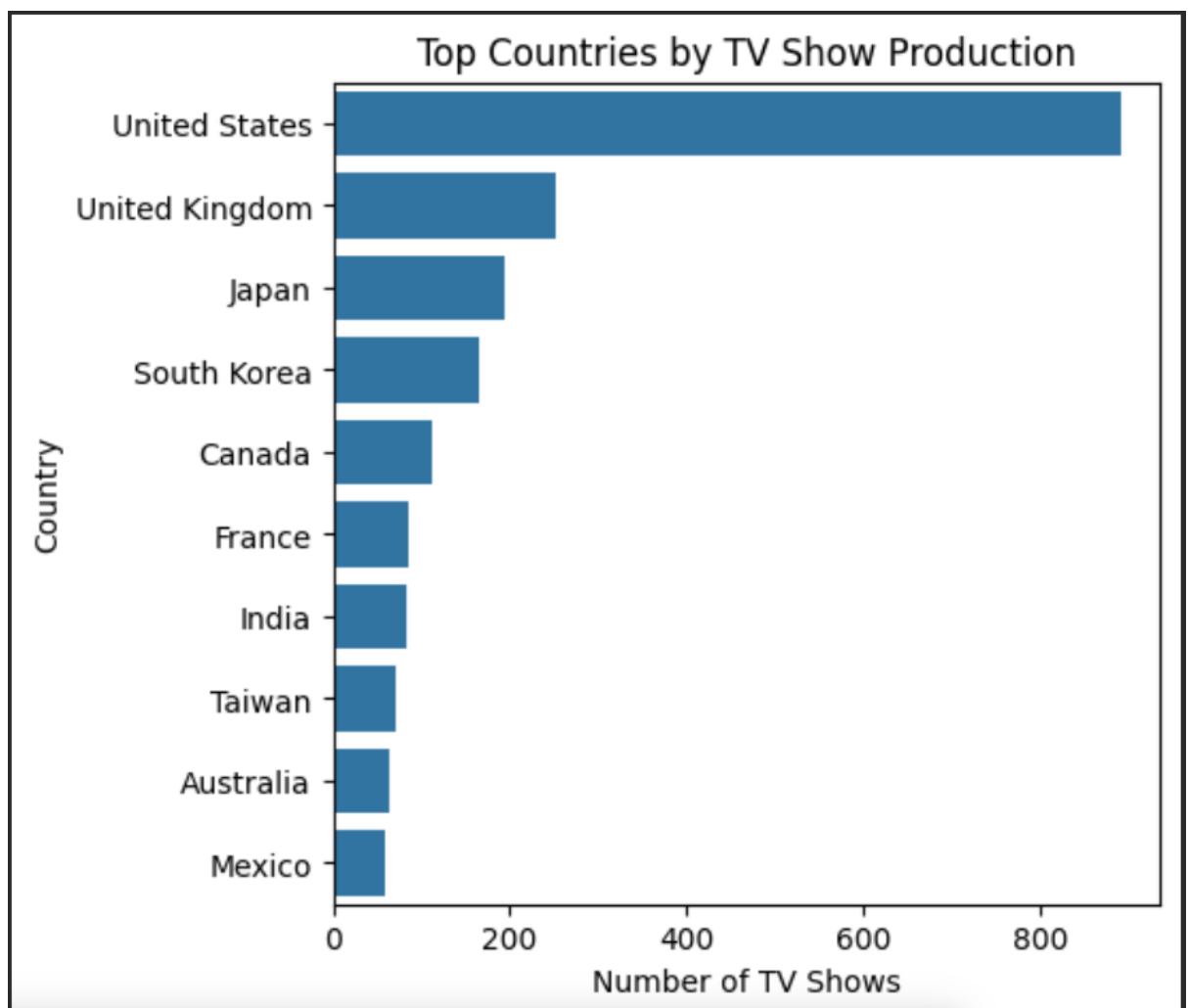


Explanation

This line graph compares the yearly trend of Movies and TV Shows on Netflix. Movies dominate in earlier years, while TV Shows show strong growth after 2017. This reflects Netflix's strategic shift from primarily movie-based content to increased investment in TV series and original shows.

20. Which country has highest proportion of TV shows?

USA leads in TV show production. Other countries contribute smaller shares. This highlights strong US dominance in series content.



Explanation

This bar chart shows the countries producing the highest number of TV shows on Netflix. The USA leads significantly, indicating strong dominance in series production. Other countries contribute comparatively smaller shares, showing that TV show production is concentrated in a few major content-producing regions.

Conclusion

The analysis reveals that Netflix initially focused more on movies but has gradually shifted toward TV shows, especially after 2017. A small number of countries dominate content production, with strong representation from the USA and India. Genres like drama and comedy remain consistently popular, indicating stable audience preferences.

The platform experienced rapid expansion between 2015 and 2020, followed by a more stable growth phase. Additionally, the dominance of mature content ratings suggests Netflix primarily targets adult audiences.

Overall, the project highlights how data analysis can uncover content strategies, user preferences, and platform evolution over time.

-----*****-----